"What is the overall correlation between vaccination rates and reduction in COVID-19 cases and deaths across top-affected countries? (2020-2023)"

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Introduction

- The COVID-19 pandemic impacted countries differently in infection rates, vaccination coverage, and mortality.
- Understanding why some countries fared better helps shape future public health preparedness.
- This project explores the role of vaccination rollout, stringency policies, and health security preparedness.
- Despite global interconnectedness, no unified exploration compares deaths, vaccination, and preparedness across major nations.
- What are the most influential factors in determining COVID-19 death rates across countries?

Materials

- JHU CSSE COVID-19 Data (GitHub)
- GovEx Global Vaccine Data (GitHub)
- GHS Index data from ghsindex.org
- Google Collab

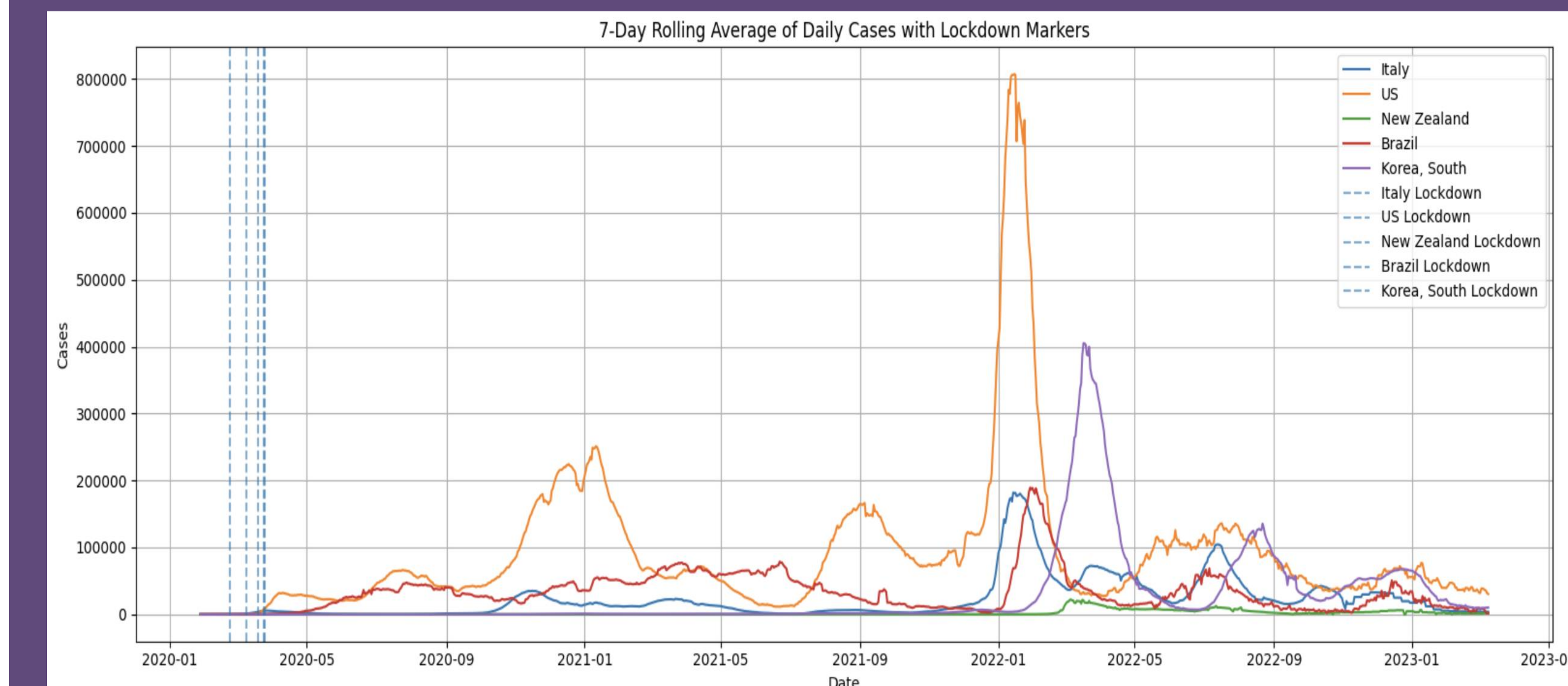
Methodology

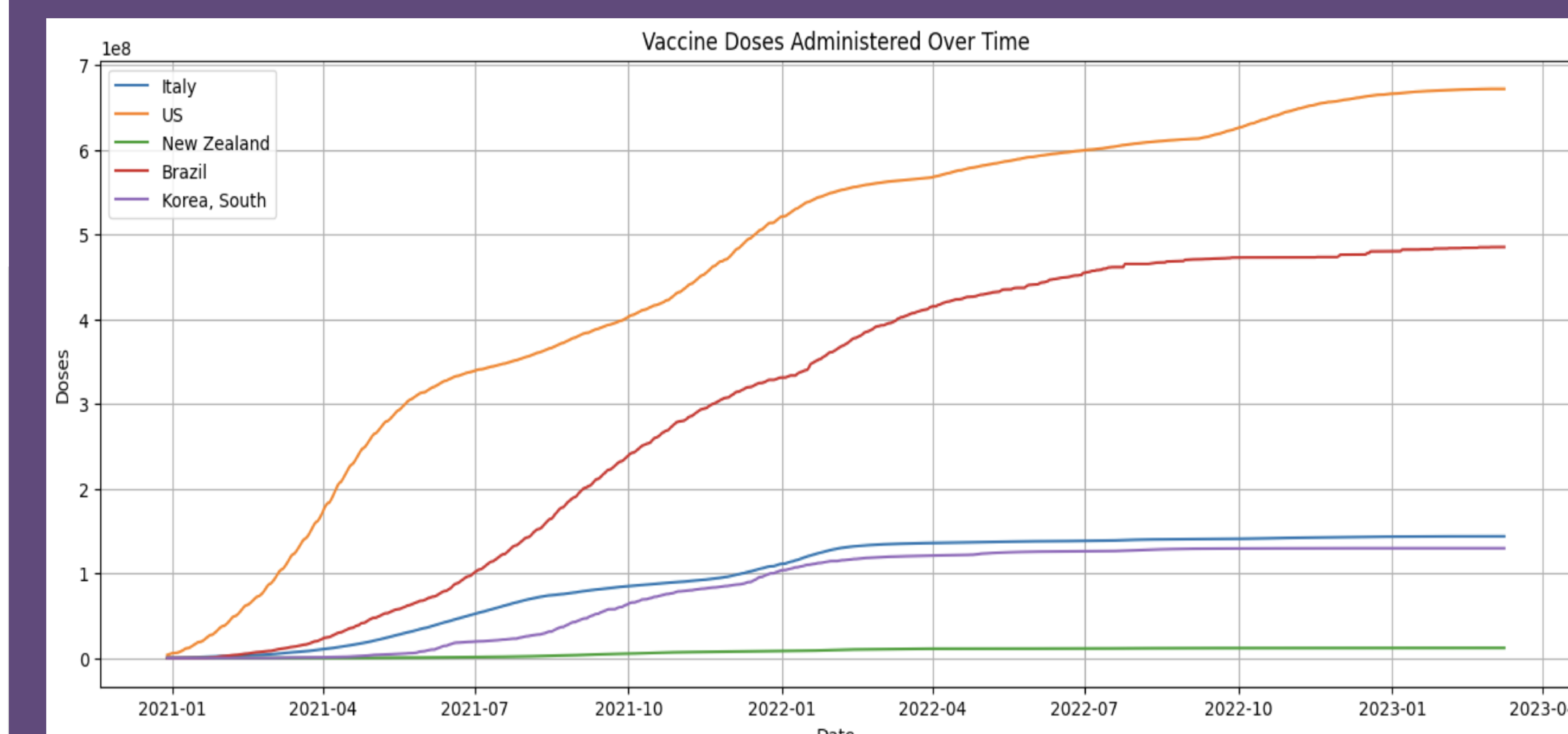
- Merged COVID-19 confirmed, death, and vaccination datasets into a unified format.
- Cleaned and normalized values as *per million (cases/deaths)* and *per hundred (vaccinations)* using population data.
- Selected 10 globally impactful countries for focused comparative analysis.

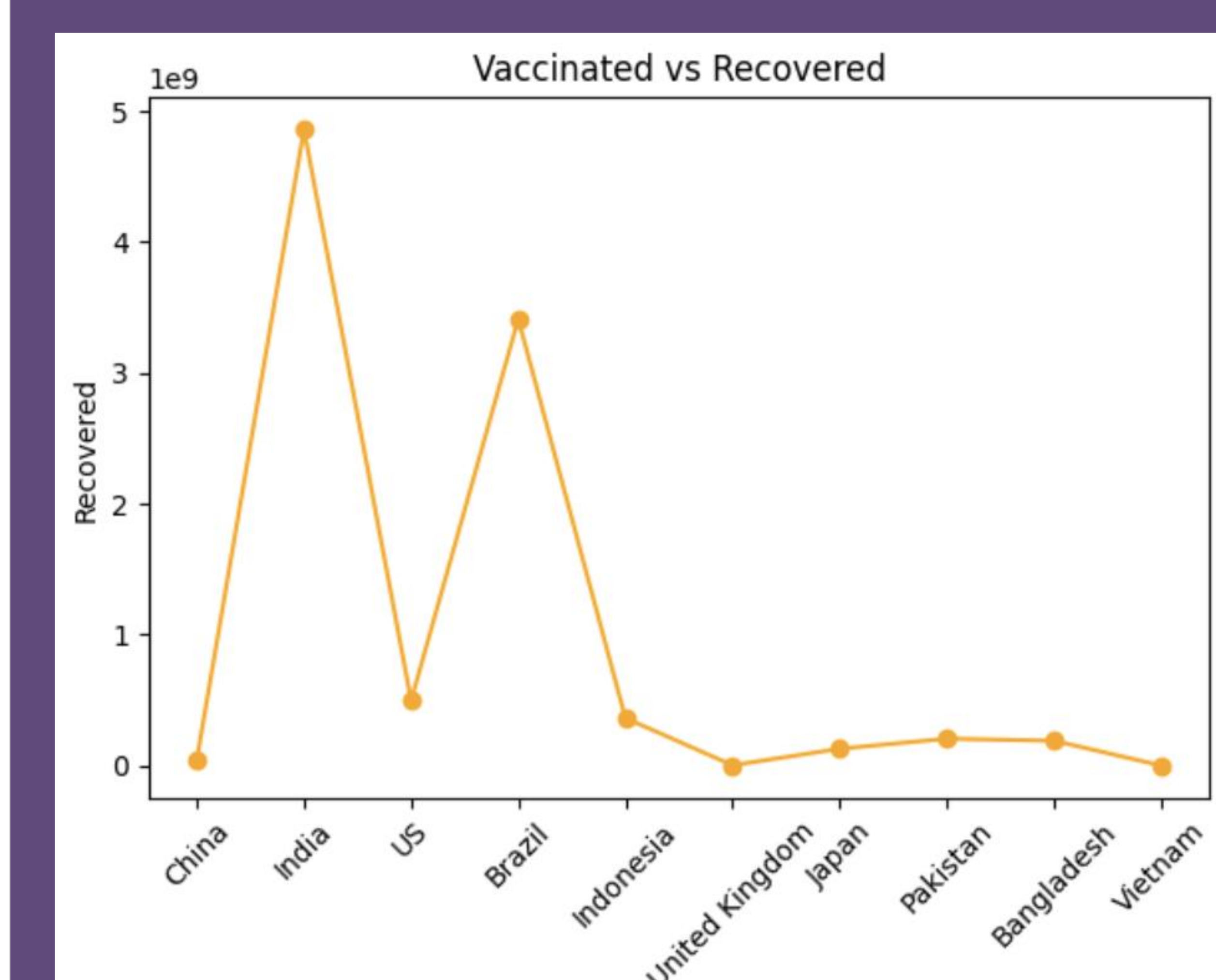
Performed:
- Correlation analysis
- Linear and KNN regression modeling
- SIR epidemiological modeling

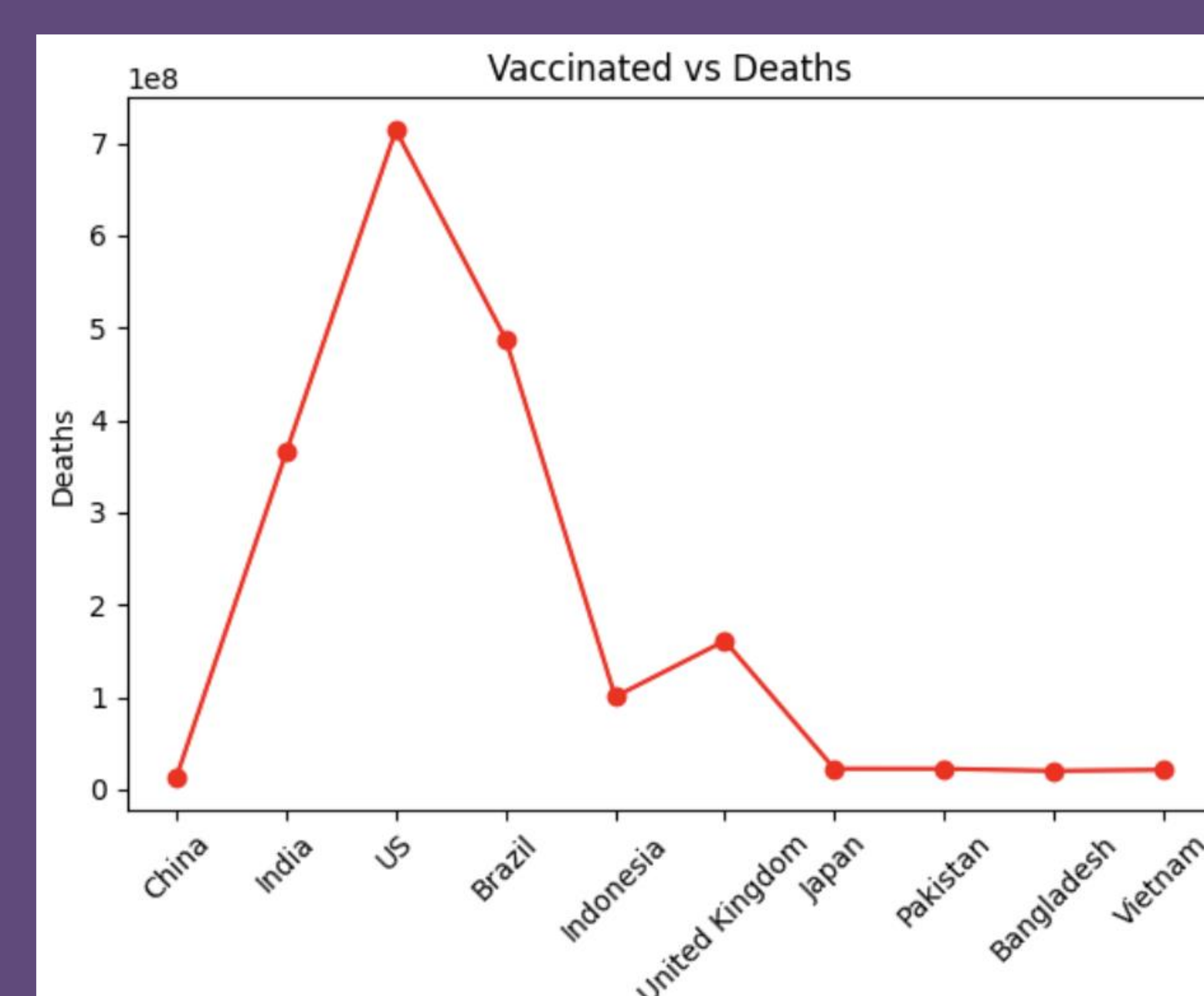
- Highlighted one country (UK or Israel) to showcase SIR model outcomes.
- Visualized trends using line plots, scatter plots, heatmaps, and bubble charts.

Results









Vaccine vs Death Correlation

- Strong inverse relationship (more vaccines, fewer deaths)
- Countries like Israel, UK show clear benefits

Deaths Over Time

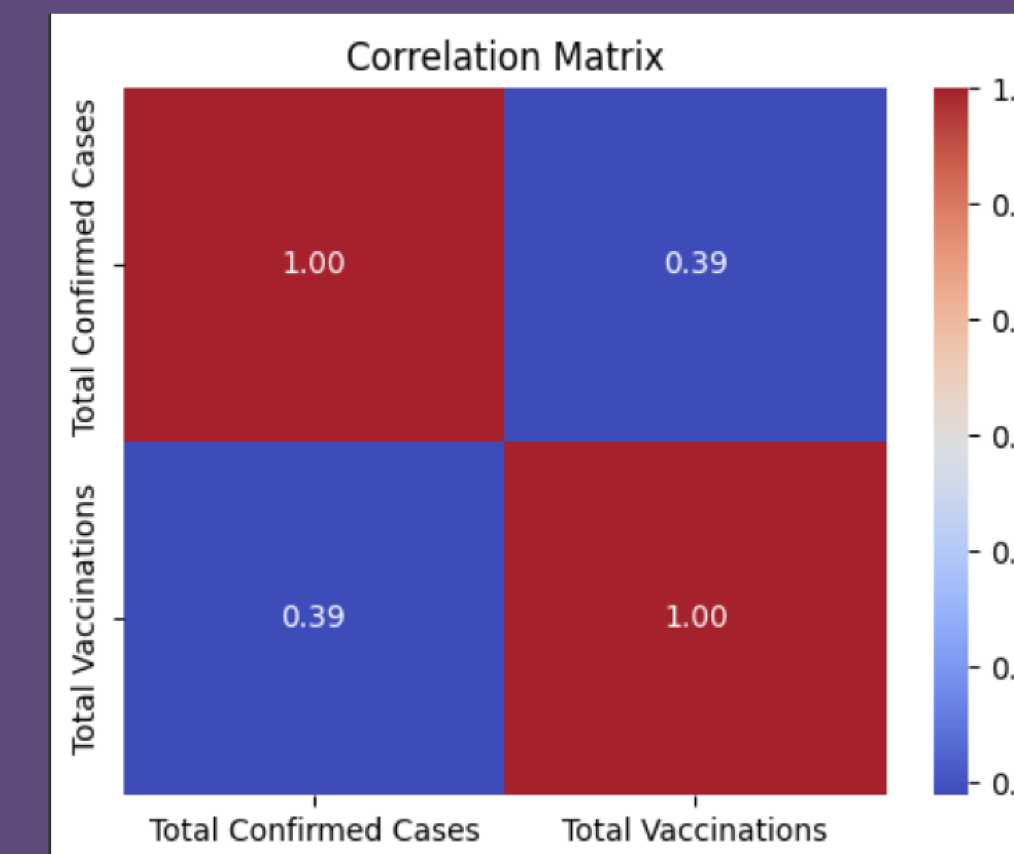
- UK & Brazil: stark contrast in post-vaccination mortality
- India's second wave visible despite early vaccinations

Correlation Matrix

- Vaccination = highest negative correlation with death rate
- Stringency Index = moderate influence

Machine Learning Models

- **Linear Regression** $R^2 \approx 0.63$
- **KNN Regression** $R^2 \approx 0.67$ (captures non-linearity)
- Key Feature: Vaccination rate



Conclusion

- Vaccination rollout is the strongest protective factor against high death rates.
- Preparedness index (GHS) doesn't always predict real-world outcomes.
- SIR model effectively simulates how public health actions alter transmission dynamics.

Future Work

Next Steps:

- Extend to newer variants (e.g., Omicron)
- Explore socioeconomic impacts
- Cluster countries using unsupervised learning
- SIR Model: UK Focus
  Infection curves flatten post-vaccinerollout
  Visual match between predicted & realinfection drop

References

- Dong E, Du H, Gardner L. An interactive web-based dashboard to track COVID-19 in real time. Lancet Inf Dis. 20(5):533-534. doi: 10.1016/S1473-3099(20)30120-1
- John Hopkins University

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